

2. Project Objective

2.1 Primary Objective of the Project

The primary objective of the **ReqRes API Testing Project using Postman** is to validate the functionality, reliability, and correctness of RESTful APIs provided by the ReqRes service (<https://reqres.in>). This is achieved by sending different types of HTTP requests and verifying whether the API responses match the expected results.

The project focuses on ensuring that each API endpoint performs its intended operation correctly, such as user authentication, retrieving user data, creating new users, updating existing users, and deleting users.

In simple terms, the main goal is to confirm that the backend APIs are working properly before they are used in any real application.

2.2 Functional Objectives

The functional objectives define what the API should do and what is being tested in this project:

1. Authentication Validation

The project verifies the login API functionality by:

- Sending valid login credentials
- Ensuring a successful response is returned
- Validating that a token is generated after login

This confirms that the authentication mechanism of the API is working correctly.

2. User Data Retrieval

The project tests APIs that fetch user data:

- Get all users
- Get a single user by ID

The objective is to ensure:

- The correct user data is returned
- The response structure is consistent
- Proper status codes are received (e.g., 200 OK)

3. User Creation

The project validates the API responsible for creating new users.

The objectives include:

- Sending valid user details (name, job, etc.)
 - Ensuring a new user is successfully created
 - Verifying that a unique user ID is generated in the response
-

4. User Update Operations

The project tests both:

- **PUT request (full update)**
- **PATCH request (partial update)**

The objective is to ensure:

- User details are correctly updated
 - API returns expected response after modification
 - Partial updates do not affect unchanged fields
-

5. User Deletion

The project validates the DELETE API to ensure:

- Users can be deleted successfully
 - The correct status code (204 No Content) is returned
 - The system handles deletion requests properly
-

2.3 Technical Objectives

The technical objectives focus on how testing is performed:

- To use **Postman** for sending API requests
- To validate **HTTP status codes** (200, 201, 204, 404, etc.)
- To verify **response body structure and data accuracy**
- To implement **test scripts using JavaScript in Postman**
- To check **response time performance**
- To organize APIs using Postman collections and environments

2.4 Automation and Execution Objectives

This project is not only manual testing but also includes automation execution:

- To execute APIs using **Postman Collection Runner**
- To automate test execution using **Newman CLI**
- To generate structured test reports after execution
- To simulate CI/CD pipeline style execution (real-world QA process)

2.5 Quality Assurance Objectives

The project ensures overall API quality by focusing on:

- Correctness of API responses
- Stability of endpoints under different inputs
- Handling of positive and negative test cases
- Data integrity between request and response
- Ensuring APIs behave consistently across multiple runs

2.6 Learning Objectives

From a learning perspective, this project helps in:

- Understanding REST API architecture
- Learning HTTP methods and status codes
- Practicing real-time API testing workflows
- Writing test scripts in Postman
- Learning automation basics using Newman
- Preparing for API testing interviews and real projects

2.7 Final Objective Summary

The overall objective of this project is to build strong hands-on experience in API testing by validating ReqRes REST APIs using Postman and Newman. It helps simulate a real QA environment where testers ensure that backend services are reliable, accurate, and ready for production use.
